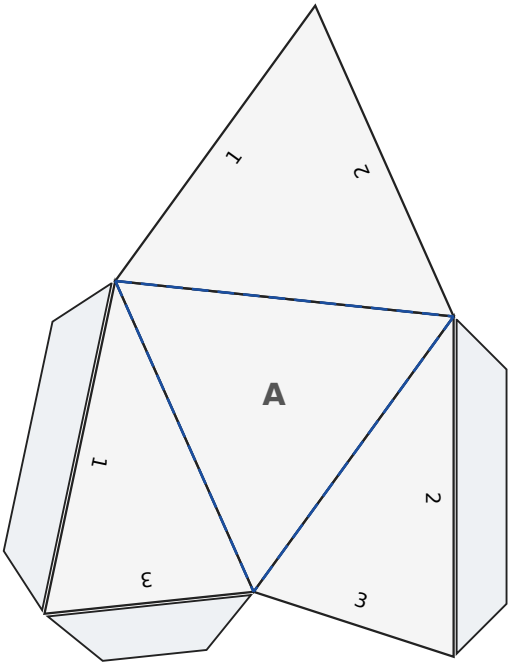
AXE (make 30; 60 for the 113-piece build)

Tetrahedron: five edges 1, one edge 1/φ. Volume 1/12. The Mosseri–Sadoc tile t6 at scale 1/τ. Fills the groove between two J11.

Cut on solid lines. Fold: dashed = mountain (crease away from you), dash-dot = valley (crease towards you); printed side out.

Glue each grey tab under the edge with the same number. Pieces are lettered A, B, ...

4 faces in 1 piece (A); 3 glue pairs; scale 45 mm per unit. Checked: no overlaps, every edge folded or paired.



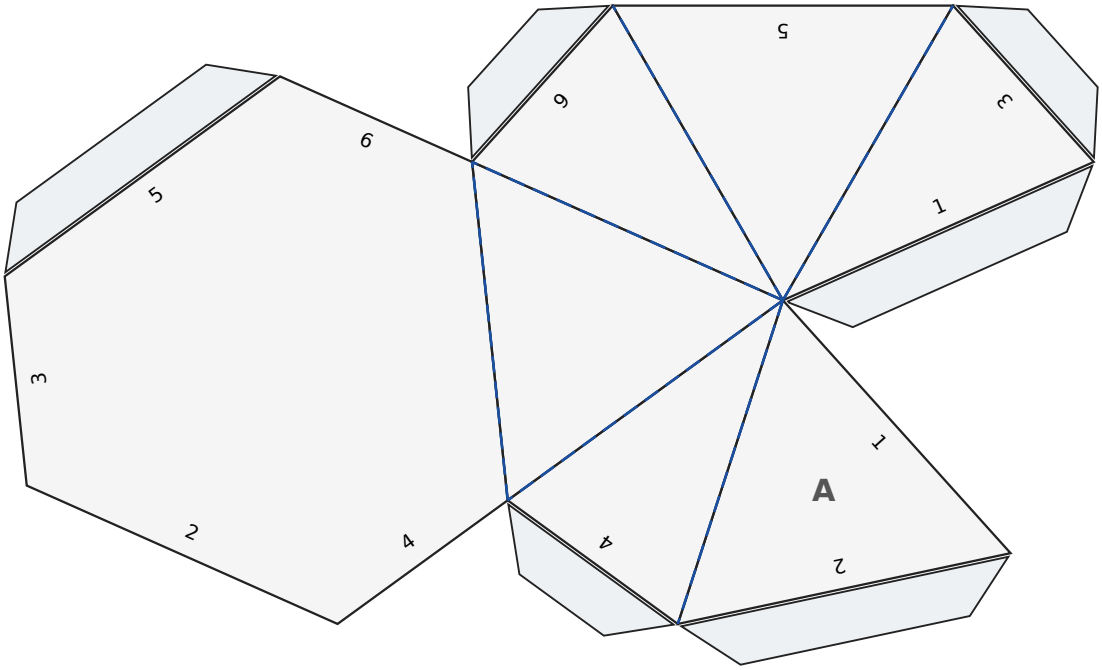
FUJI (make 20)

Hexagonal pyramid: base equiangular, sides 1 and 1/φ alternating; height 1/√3. Volume (1 + 3√5)/24. Fills each dodecahedron c

Cut on solid lines. Fold: dashed = mountain (crease away from you), dash-dot = valley (crease towards you); printed side out.

Glue each grey tab under the edge with the same number. Pieces are lettered A, B, ...

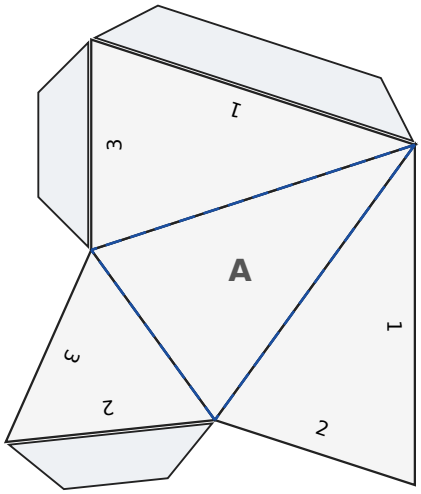
7 faces in 1 piece (A); 6 glue pairs; scale 45 mm per unit. Checked: no overlaps, every edge folded or paired.



CLEO (make 20, for the 113-piece build and the star solids

Triangular pyramid: base equilateral 1/φ, side edges 1. Volume (√5 − 1)/24. The Mosseri–Sadoc tile t4 at scale 1/τ.

Cut on solid lines. Fold: dashed = mountain (crease away from you), dash-dot = valley (crease towards you); printed side out.
Glue each grey tab under the edge with the same number. Pieces are lettered A, B, ...
4 faces in 1 piece (A); 3 glue pairs; scale 45 mm per unit. Checked: no overlaps, every edge folded or paired.



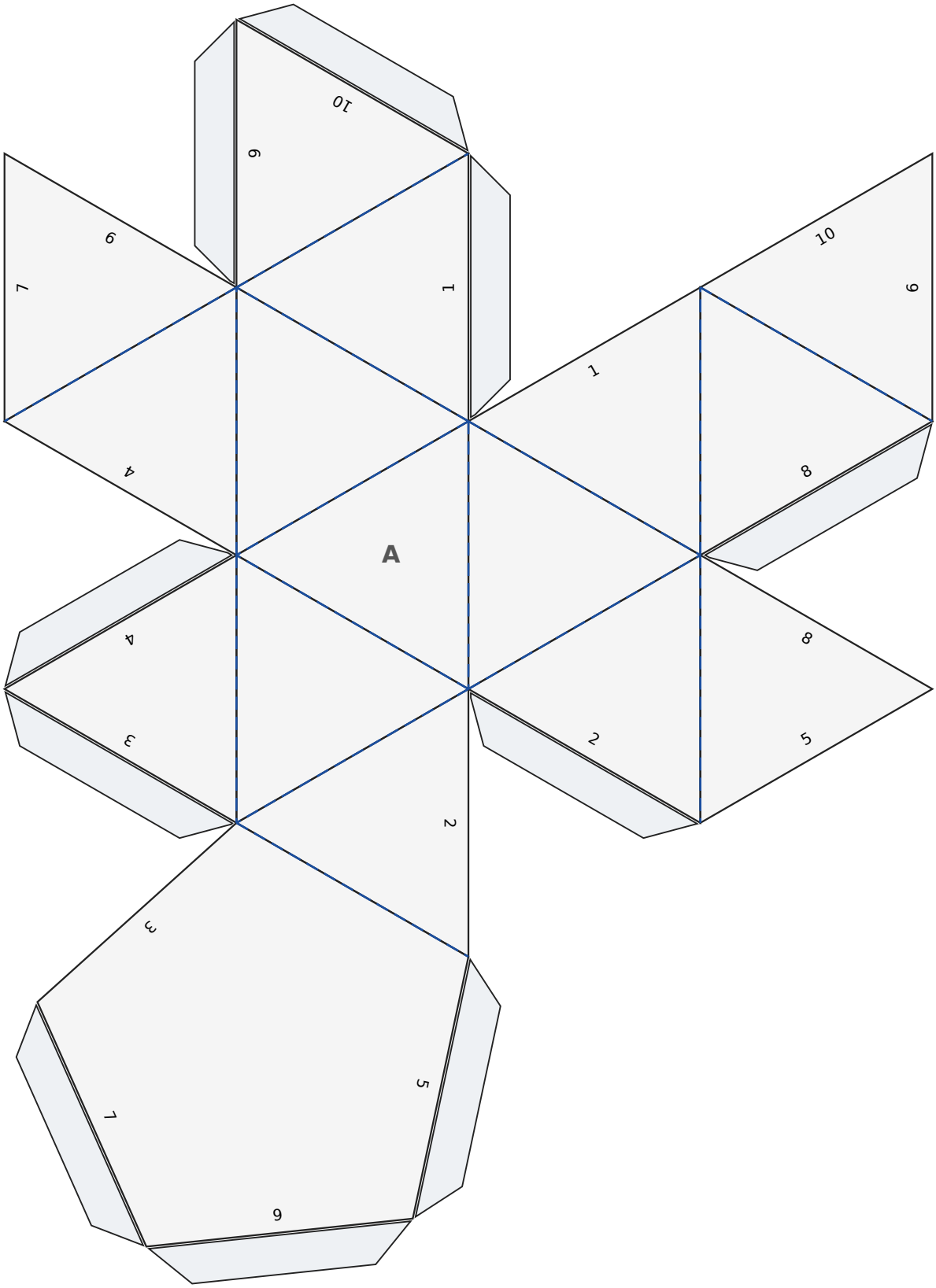
J11, gyroelongated pentagonal pyramid (make 12)

Icosahedron less one pentagonal-pyramid cap (Johnson solid J11). Its pentagon sits on a dodecahedron face.

Cut on solid lines. Fold: dashed = mountain (crease away from you), dash-dot = valley (crease towards you); printed side out.

Glue each grey tab under the edge with the same number. Pieces are lettered A, B, ...

16 faces in 1 piece (A); 10 glue pairs; scale 45 mm per unit. Checked: no overlaps, every edge folded or paired.



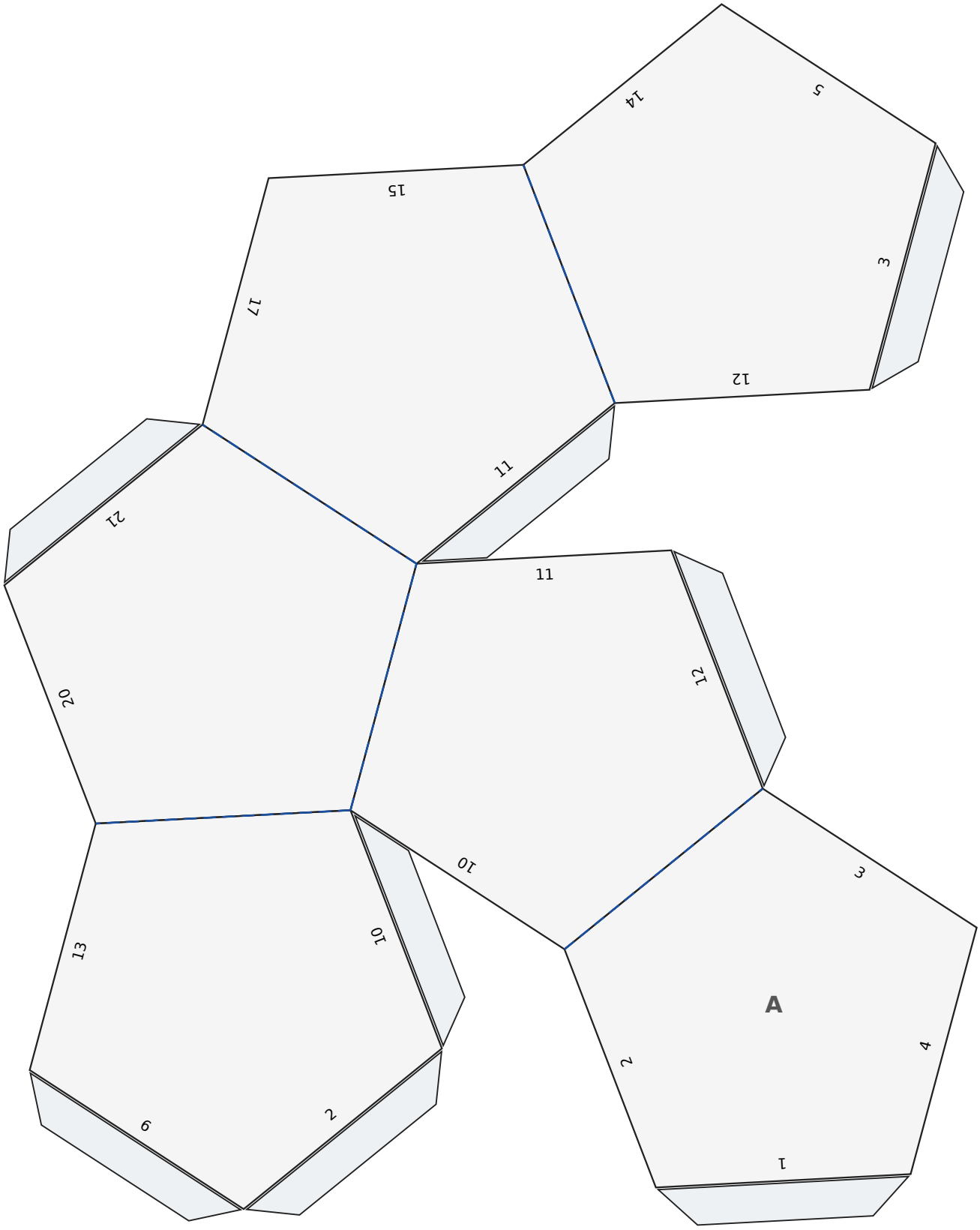
Dodecahedron, edge 1 (make 1)

The core. With 12 J11, 30 AXE and 20 FUJI it makes the Lotus seed: an icosahedron of edge φ², 63 pieces.

Cut on solid lines. Fold: dashed = mountain (crease away from you), dash-dot = valley (crease towards you); printed side out.

Glue each grey tab under the edge with the same number. Pieces are lettered A, B, ...

12 faces in 3 pieces (A, B, C); 21 glue pairs; scale 45 mm per unit. Checked: no overlaps, every edge folded or paired.



Dodecahedron, edge 1 (make 1)

The core. With 12 J11, 30 AXE and 20 FUJI it makes the Lotus seed: an icosahedron of edge φ², 63 pieces.

Cut on solid lines. Fold: dashed = mountain (crease away from you), dash-dot = valley (crease towards you); printed side out.
Glue each grey tab under the edge with the same number. Pieces are lettered A, B, ...
12 faces in 3 pieces (A, B, C); 21 glue pairs; scale 45 mm per unit. Checked: no overlaps, every edge folded or paired.

